
accelerate
collaborate
innovate
communicate
palm
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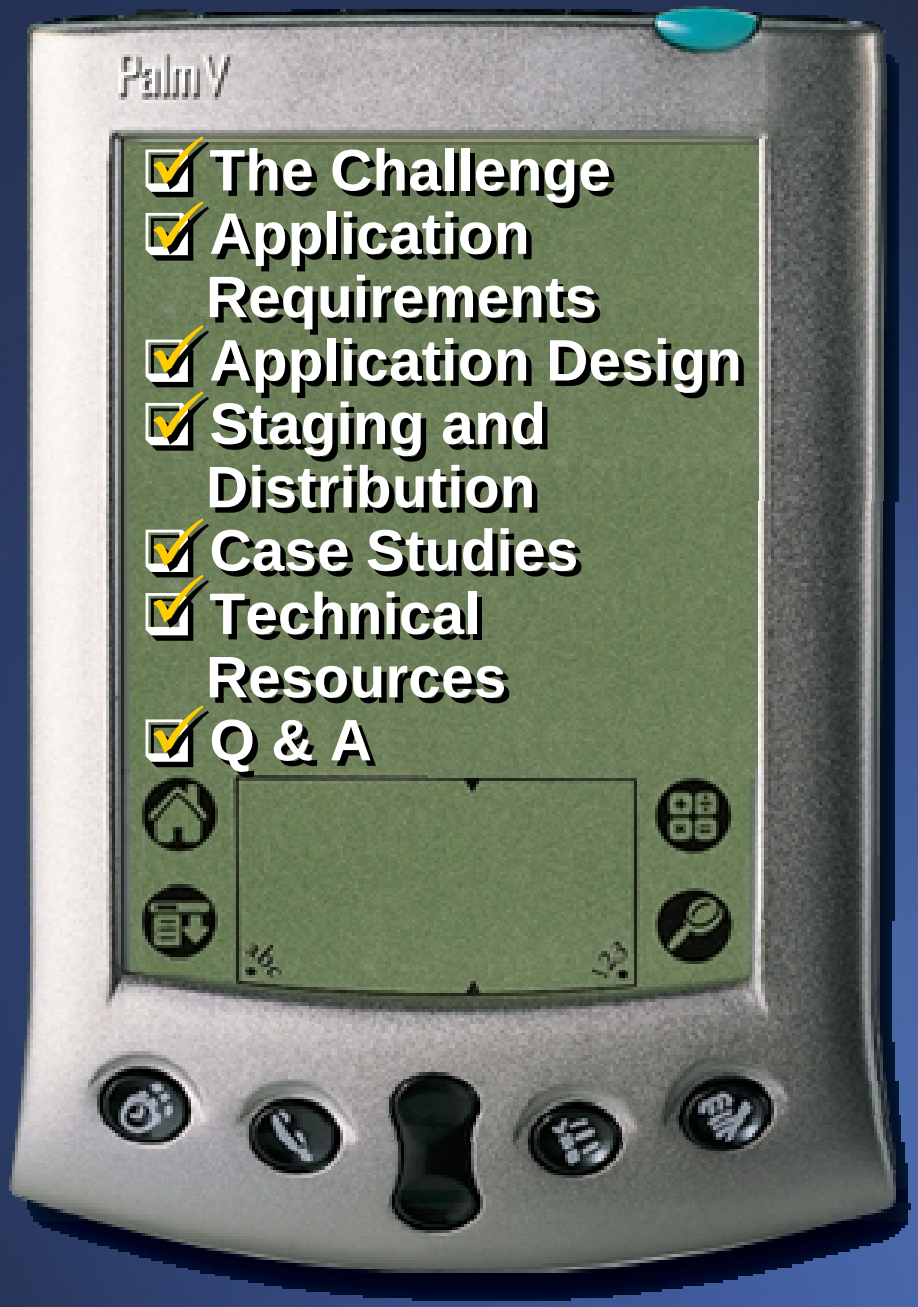


How to Develop and Design for the Enterprise

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Software Product Manager

PalmSource
Conference 2000

Agenda


Palm V
The Challenge
Application Requirements
Application Design
Staging and Distribution
Case Studies
Technical Resources
Q & A

The Challenge

- 5000 users → 10,000
- Immediate access to corporate data
- Must work with existing systems
- Encrypted, password protected
- Wireless / Wireline
- Fast, efficient, easy to use
- Need it tomorrow

The Pieces Are in Place

- Palm Synchronization to…
  - Oracle
  - Sybase
  - DB2
  - SAP
  - Notes
  - Web
- Hardware choices
- The ROIs are large
  - Paper processes are inefficient
    - Printed data is old
    - Requires manual re-entry
    - Lost data

Application Requirements

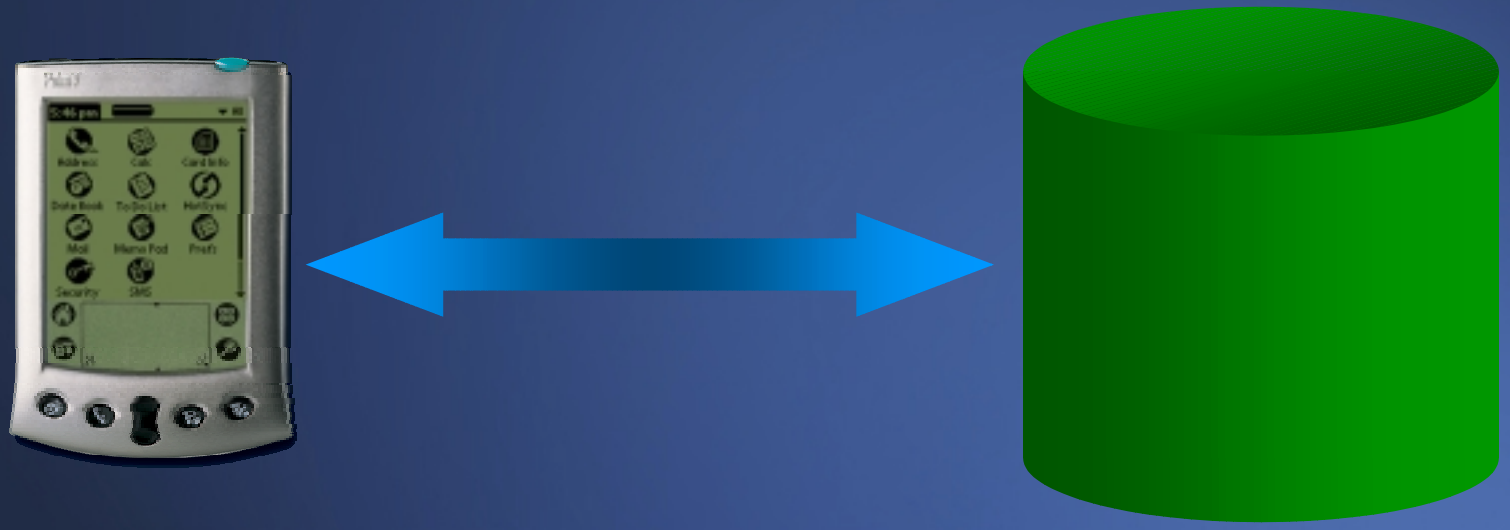
- How will the users synchronize?
- Partitioning an Application
- Security
- Data entry options
- Hardware choices

Application Requirements
How Will the Users Synchronize?

- Batch mode vs. real time
  - Timeliness of Data?
  - $$ and complexity
- Connectivity
  - Palm Cradle
  - Analog Modem
  - Wireless LAN
  - Wireless WAN
- Mobile vs. Stationary

Application Requirements
Partitioning an Application

- Local storage vs. server storage
- What portion synchronizes with Palm OS™
  - User specific content



Application Requirements
Security

- Device security
- Data encryption
  - Device databases
  - Over the air
- User authentication

Medical Records, Stock Transactions, Gov. Data?

Application Requirements
Data Entry

- Maximize check boxes and pick lists
- Efficient data entry
  - Bar Code
  - Mag Stripe
  - Infrared
  - Data Acquisition
  - Snap on Keyboard

Application Requirements
Hardware Peripherals

- Rubber boot
- Analog / Wireless modem
- Infrared printer
- GPS
- Vehicle mounts
- Ethernet cradle / IR device
- Bar Code wand

Application Design

- 3 Development architectures
  - Forms-based
  - C / C++
  - Web
- Middleware for enterprise systems
- Server synchronization
- 3 Synchronization architectures
  - Local
  - Server synchronization
  - Web based synchronization

Application Design
Forms-Based

- Pros
  - Rapid development
  - Device side control / logic
  - Minimal conduit development
  - Customizable
- Cons
  - Run time fees
  - Some UI restrictions

Application Design
C / C++

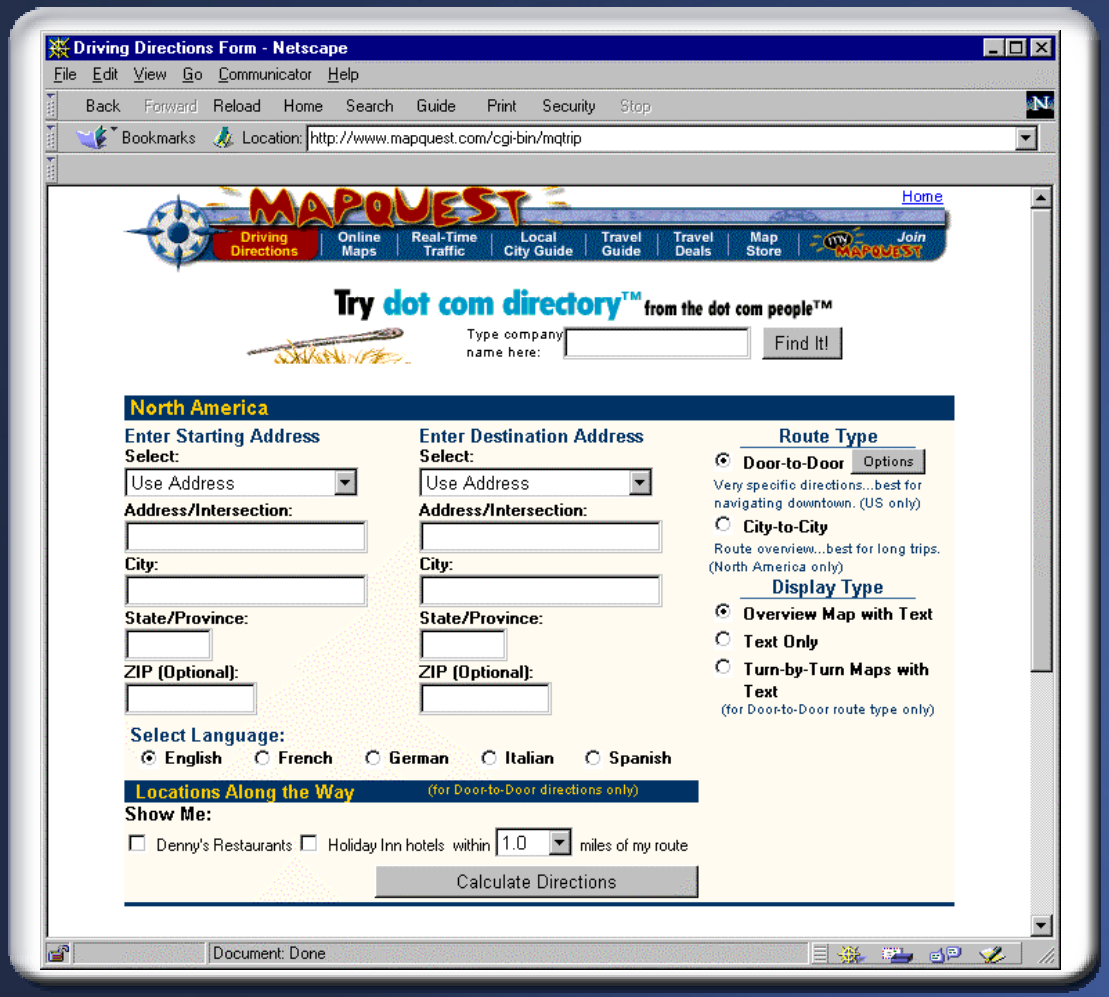
- Pros
  - Efficient / fast application
  - Large pool of experienced developers
    - >50,000 registered with Codewarrior
  - No run time fees
- Cons
  - Conduit development required
  - Experienced programmer required

Application Design
Web

- Pros
  - No conduit development required
  - Development in HTML
  - Use existing web pages
  - Supports batch and real-time modes
  - Easily distribute application changes
- Cons
  - Limited device side logic

Application Design

Make the Web Palm OS Friendly





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Application Design
Solutions Exist for Enterprise Data

- Oracle
  - Oracle Lite
- Sybase
  - Sybase Ultralite
- DB2
  - DB2 Everyplace
- Legacy (OS400, Terminal emulator)
  - Attachmate, Translink
- Web
  - AvantGo, Palm.Net®, Browse-it

Application Design
Server Synchronization Tools

- Palm HotSync® Server
- AvantGo
- Synchrologic
- Aether ScoutSync
- Intellisync Anywhere
- IBM Mobile Connect
- XTND-Connect

Application Design

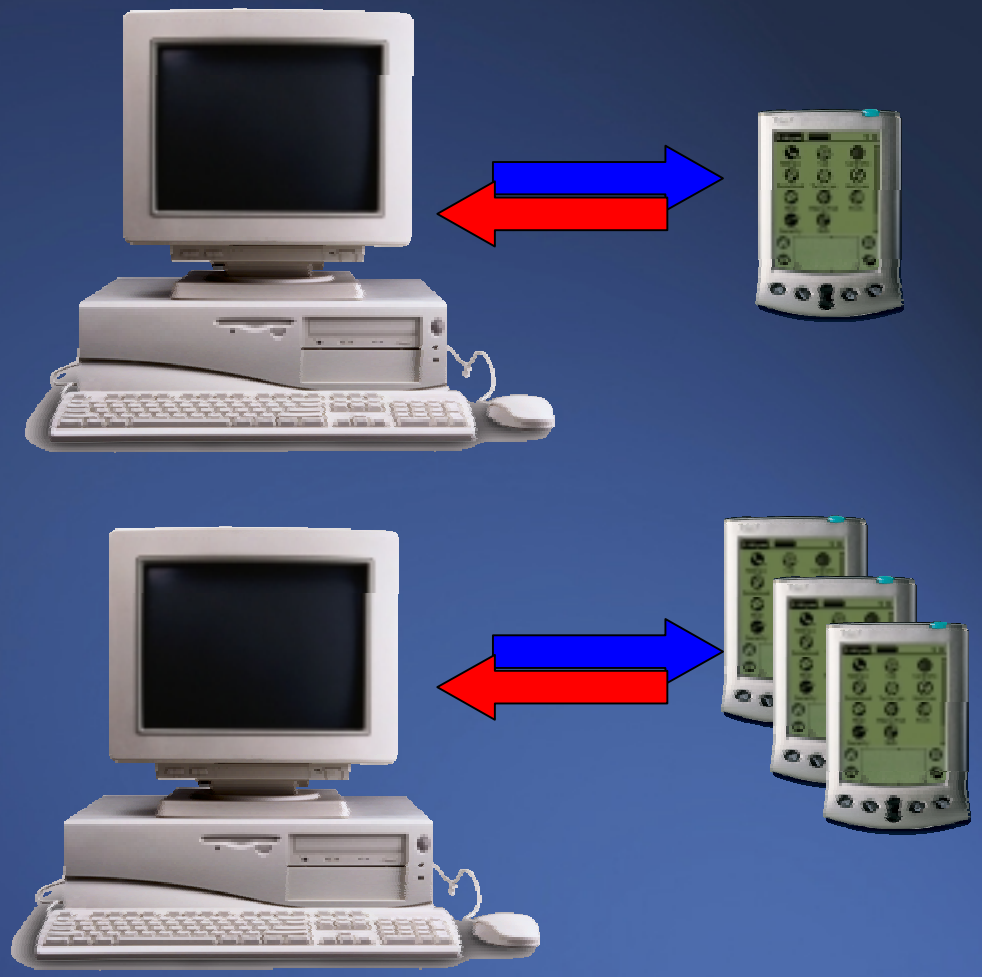
4 Network Architectures

- Local PC synchronization
- Server synchronization
- Web Server synchronization
- Palm VII / Palm.net

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Application Design
Network Architecture #1

- Local PC Synchronization
  - One synchronization at a time
  - Small PC application
  - Batch mode
  - non-wireless



Application Design
Network Architecture #2

- Server Synchronization
  - Simultaneous Synchronization
  - Scalable
    - 1000's of users
    - Multiple Palm OS devices
  - Networked Environment
  - Secure
  - Batch mode



Application Design
Network Architecture #3

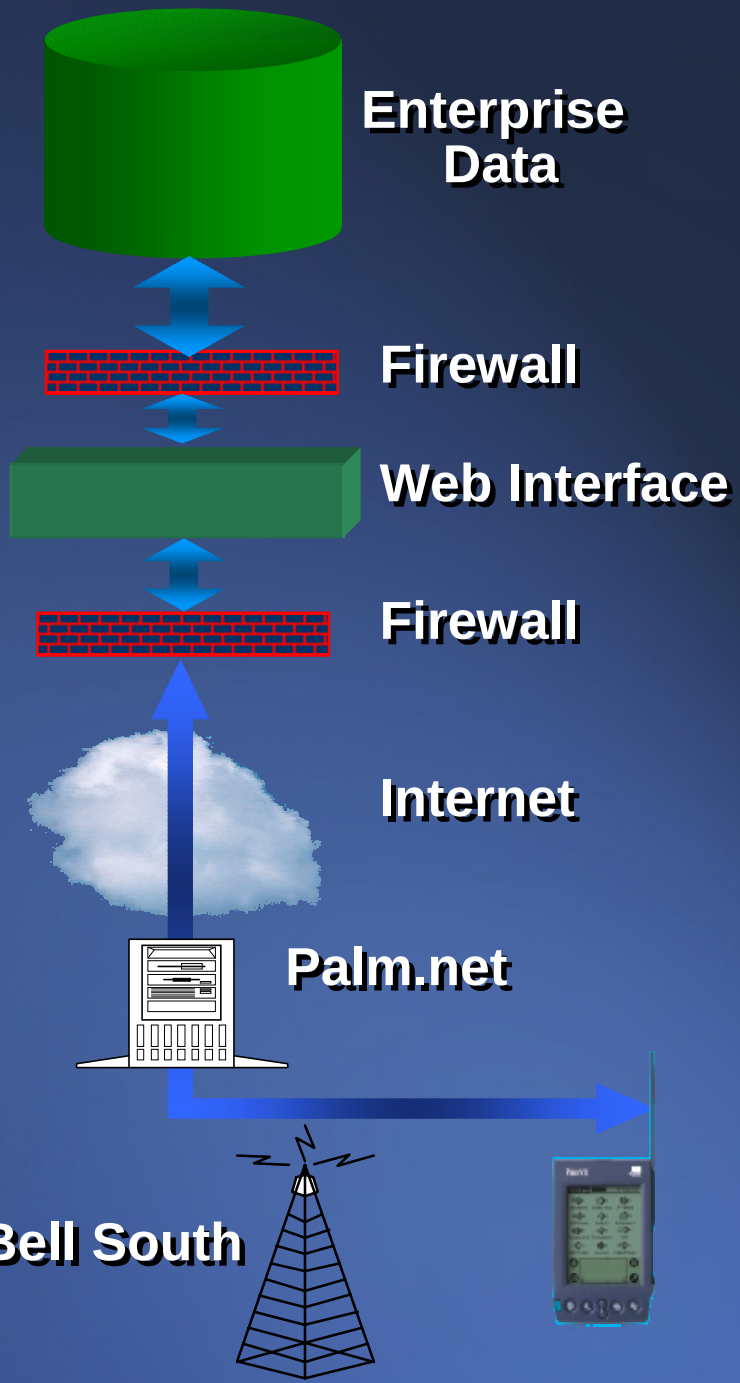
- Web Synchronization
  - AvantGo
  - Pumatech Browse-it

  - Leverage Intranet investment
  - Batch or real time



Application Design
Network Architecture #4

- Palm VII / Palm.Net
  - Real Time
    - Query based
    - Request / Response
  - Secure
  - Minimal Development



Application Distribution and Staging

- Building a device image
- Application distribution

Application Distribution and Staging Building a Device Image

- How to Image 5000 devices??
  - 1) Build the perfect Palm OS handheld
  - 2) Backup buddy - obtain every prc/pdb from the device
  - 3) Setup Sync stations, place prcs/pdbs in backup folder
  - 4) Sync til you Drop
  - 5) Remove the username??

- With applications in Flash
  - TRG InstallPro

Application Distribution and Staging Application Distribution

- Tivoli
- Computer Associates
  - Ship IT
  - Aim IT
- Using Server Synchronizaiton
- Email new prcs??

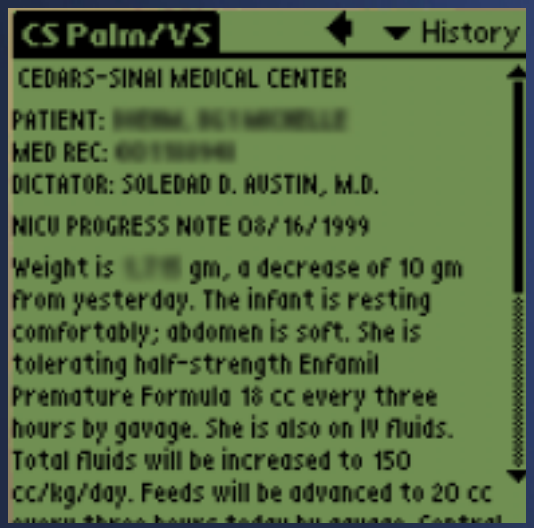
Case Study #1
LA City Fire Inspection

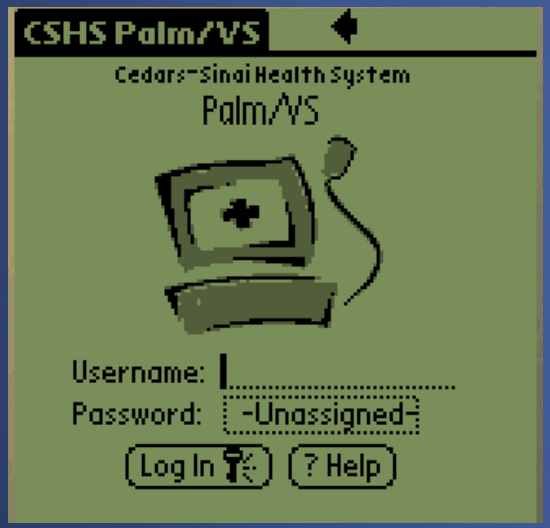
- Forms-based application
- Local Synchronization
- Efficient data entry
- Batch mode
- Specifics
  - MS Access Database
  - Spmbol 1500
  - Satellite Forms
- ROI

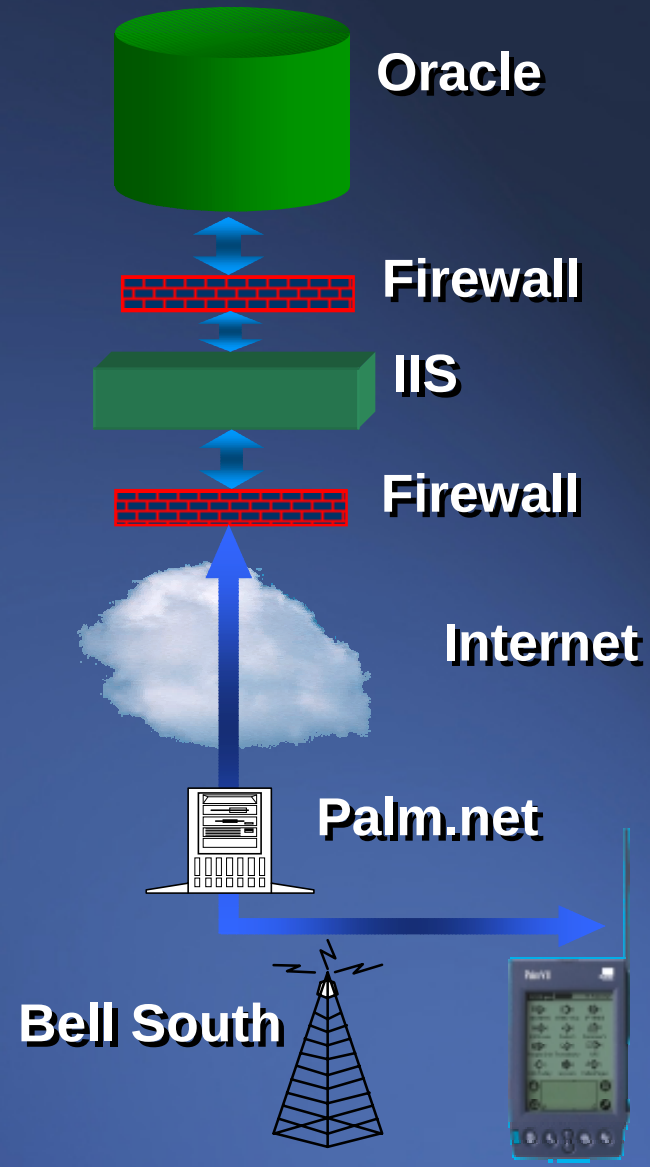


Case Study #2
Cedars-Sinai Palm/VS

- Doctors can query
  - Transcribed reports
  - Lab data
- Secure
- Real time

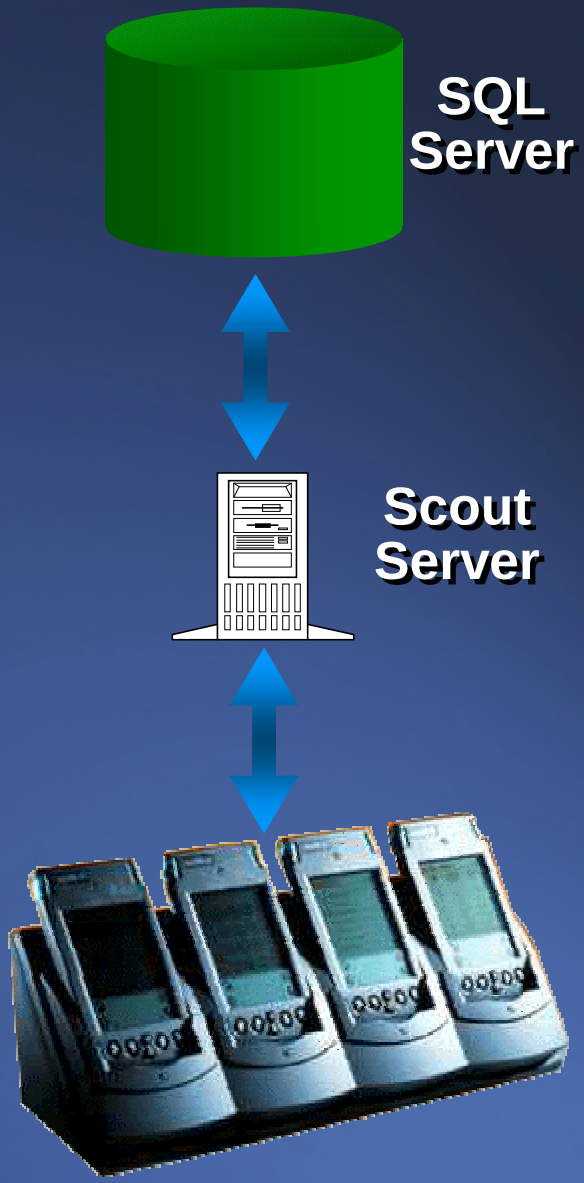






Case Study #3
US Postal Office

- Codewarrior C / C++ application
- 12 airports each with a Symbol Ethernet cradle
- 1 Scout Server
- Scalable



Technical Resources

- Palm Computing resources
- Third party resources


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